

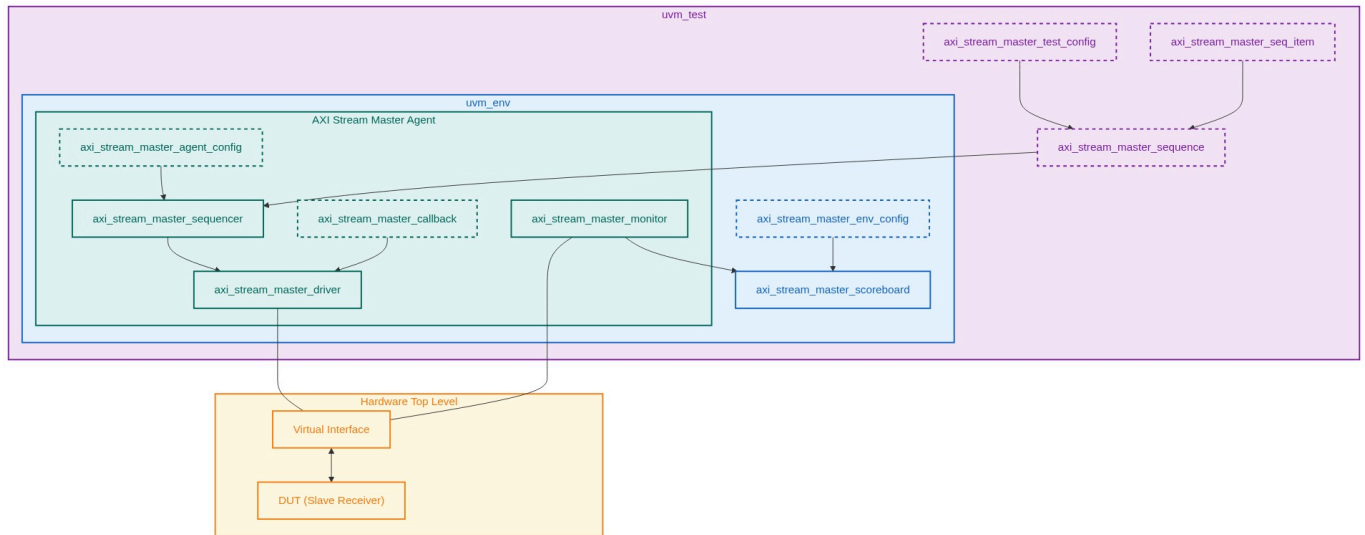
Verification Plan Report

Project: AXI5-Stream Protocol Verification

Version: 1.4

Verification Role: Master

1. UVM Environment Architecture



2. Test Requirements

Index	Feature/Sub	Name	Description
REQ_MST_0 1	Handshake (TVALID Stability)	TVALID Persistence Until Handshake	The AXI-Stream protocol mandates that once TVALID is asserted by the Master, it must remain HIGH until a successful handshake (TVALID & TREADY) occurs. This prevents the loss of data or corruption of the stream state. The VIP Master Driver must be configured to assert TVALID and maintain it regardless of TREADY backpressure. The monitor must flag a protocol violation if TVALID drops without TREADY being asserted.
REQ_MST_0 2	Handshake (Flow Control)	Two-way Flow Control Handshake	Verification of the bidirectional flow control mechanism where either TVALID or TREADY can be asserted first. The protocol allows for zero-wait-state transfers where both are asserted in the same cycle. Test cases must cover all three handshake scenarios: TVALID before TREADY, TREADY before TVALID, and simultaneous assertion. This ensures the Master correctly interprets the Slave's readiness.

Verification Plan Report

REQ_MST_0 3	Data Transfer (TDATA Integrity)	Payload Data Integrity	The primary payload TDATA must be correctly transmitted from the Master to the Slave without corruption. The width of TDATA must be an integer number of bytes. The scoreboard must compare the data driven by the Master VIP against the data sampled at the Slave interface, ensuring bit-accurate transfer across all byte lanes.
REQ_MST_0 4	Byte Qualification (TKEEP/TSTRB Logic)	Byte Lane Qualification and Null Bytes	TKEEP and TSTRB signals qualify the content of TDATA. TKEEP indicates if a byte is a null byte (to be removed), while TSTRB distinguishes between data and position bytes. Verification must ensure that the Master correctly drives these qualifiers to support sparse streams and unaligned packet starts/ends. Illegal combinations (TKEEP=LOW, TSTRB=HIGH) must be checked.
REQ_MST_0 5	Packet Boundaries (TLAST Signaling)	Packet Demarcation with TLAST	TLAST is used to indicate the end of a packet or frame. It allows interconnects to perform arbitration and buffer management efficiently. Test scenarios must verify that the Master correctly asserts TLAST at packet boundaries. Negative tests should include "infinite" packets where TLAST is never pulsed to check Slave overflow handling.
REQ_MST_0 6	Source/Destination (TID/TDEST Routing)	Stream and Routing Identifiers	TID identifies the stream source, and TDEST provides routing information. These signals allow multiple logical streams to be interleaved on a single physical link. The verification environment must verify that the Master can switch TID/TDEST between transfers and that the Slave correctly demultiplexes or routes the data based on these IDs.
REQ_MST_0 7	Sideband (TUSER Information)	User-defined Sideband Integrity	TUSER provides a mechanism for transport-layer metadata. It is recommended to be an integer multiple of the data width in bytes. The VIP must verify that TUSER bits are correctly associated with their respective bytes and that trimming/padding rules are followed if the bus width changes in an interconnect.
REQ_MST_0 8	AXI5 Features (TWAKEUP Signaling)	Low Power Wakeup Handshake	AXI5-Stream introduced TWAKEUP to indicate interface activity. It must be glitch-free and can be asserted before TVALID. Verification must ensure the Master asserts TWAKEUP to exit low-power states before initiating transfers and maintains it until the transfer completes.
REQ_MST_0 9	AXI5 Features (Parity Protection)	Interface Parity Error Detection	AXI5 parity (TDATACHK, TVALIDCHK, etc.) uses odd parity over 8-bit groups to detect single-bit flips. The Master VIP must correctly generate parity bits for all signals. Negative testing must inject parity errors by flipping bits in the CHK signals to verify Slave error detection.

Verification Plan Report

REQ_MST_1 0	System (Reset Compliance)	Synchronous Reset and TVALID Gating	During reset (ARESETn LOW), TVALID must be driven LOW. All other signals have indeterminate values. TVALID remains LOW until the first ACLK edge after ARESETn deassertion. The VIP must verify that no spurious transfers occur during reset and that the Master starts the handshake protocol correctly immediately after reset is released.
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3. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_MST_001	VIP Master drives a single-beat transfer with all byte lanes enabled (TKEEP=all 1s, TSTRB=all 1s, TLAST=1).	REQ_MST_02, REQ_MST_03, REQ_MST_05
TC_MST_002	VIP Master drives a burst of 10 transfers while the Slave (DUT) holds TREADY HIGH, verifying max throughput.	REQ_MST_02, REQ_MST_03
TC_MST_003	VIP Master asserts TVALID and waits 5 clock cycles before Slave asserts TREADY, verifying REQ_MST_01 stability.	REQ_MST_01, REQ_MST_02
TC_MST_004	Slave (DUT) asserts TREADY before Master drives TVALID, verifying the Master performs a zero-wait-state handshake.	REQ_MST_02
TC_MST_005	VIP Master drives random TDATA values for 100 transfers and ensures the scoreboard confirms data integrity.	REQ_MST_03
TC_MST_006	VIP Master drives a 'Null Packet' where TVALID=1, TLAST=1, but all TKEEP bits are 0. Verify Slave ignores data.	REQ_MST_04
TC_MST_007	VIP Master drives a Sparse Stream with interleaved data and position bytes (TSTRB toggling).	REQ_MST_04
TC_MST_008	VIP Master drives back-to-back packets with zero idle cycles between TLAST=1 and the next TVALID=1.	REQ_MST_05
TC_MST_009	VIP Master interleaves two streams by rapidly switching TID (e.g., Stream A, then B, then A).	REQ_MST_06
TC_MST_010	VIP Master drives packets to multiple destinations by varying TDEST values.	REQ_MST_06
TC_MST_011	VIP Master drives randomized TUSER sideband values for every beat and verifies receipt.	REQ_MST_07
TC_MST_012	VIP Master asserts TWAKEUP, waits 3 cycles for 'wake latency', then drives TVALID.	REQ_MST_08
TC_MST_013	VIP Master drives valid transfers while asserting correct parity on TDATACHK (Odd Parity).	REQ_MST_09
TC_MST_014	Negative: VIP Master injects a single-bit error on TDATA and expects Slave to flag parity error (if supported).	REQ_MST_09

Verification Plan Report

TC_MST_015	VIP Master deasserts ARESETn and drives TVALID on the very next rising edge of ACLK.	REQ_MST_10
TC_MST_016	VIP Master performs a transfer where TREADY is deasserted mid-packet. Master must hold TVALID and TDATA.	REQ_MST_01, REQ_MST_02
TC_MST_017	VIP Master drives unaligned packet start (first beat TKEEP is not all 1s).	REQ_MST_04
TC_MST_018	VIP Master drives unaligned packet end (final beat TKEEP is not all 1s).	REQ_MST_04
TC_MST_019	VIP Master drives a single-beat packet (TLAST=1) where all bytes are position bytes (TSTRB=0).	REQ_MST_04, REQ_MST_05
TC_MST_020	VIP Master drives 5000 transfers with randomized TREADY wait states to stress handshake logic.	REQ_MST_02
TC_MST_021	VIP Master asserts ARESETn in the middle of a multi-beat packet. TVALID must immediately drop.	REQ_MST_10
TC_MST_022	VIP Master drives TUSER bits with TKEEP=LOW. Verify TUSER is correctly masked or ignored by interconnect.	REQ_MST_07
TC_MST_023	VIP Master drives a frame consisting of exactly 8 packets. Master preserves TLAST assertions.	REQ_MST_05
TC_MST_024	VIP Master drives TWAKEUP and TVALID simultaneously, ensuring TWAKEUP stays high until handshake.	REQ_MST_08
TC_MST_025	Negative: Master VIP drops TVALID before TREADY is asserted. Monitor must catch this protocol violation.	REQ_MST_01
TC_MST_026	VIP Master drives maximum TID width (e.g., 8-bit ID) and verifies ID isolation.	REQ_MST_06
TC_MST_027	VIP Master drives packets with random TLAST duration (from 1 beat to 1024 beats).	REQ_MST_05
TC_MST_028	VIP Master drives TDATA with odd number of set bits and ensures TDATACHK is LOW (Odd Parity check).	REQ_MST_09
TC_MST_029	VIP Master drives TDATA with even number of set bits and ensures TDATACHK is HIGH (Odd Parity check).	REQ_MST_09
TC_MST_030	VIP Master drives TWAKEUP then deasserts it WITHOUT driving TVALID (permitted but not recommended).	REQ_MST_08
TC_MST_031	VIP Master drives 64-bit wide TDATA and verifies TSTRB/TKEEP mapping to each byte lane.	REQ_MST_04
TC_MST_032	VIP Master drives transfers with TID/TDEST change only when TLAST=1 (Continuous Packet mode simulation).	REQ_MST_05, REQ_MST_06

Verification Plan Report

4. Functional Coverage Groups

Group Name	Description
cg_handshake	Cross of TVALID and TREADY to cover all handshake scenarios.
cg_data_width	Coverage of TDATA values across all byte lanes, including all zero and all one toggling.
cg_qualifiers	Coverage of all valid TKEEP and TSTRB bitmask combinations.
cg_packet_boundary	Coverage of TLAST transitions and packet lengths.
cg_stream_ids	Coverage of TID and TDEST ranges to ensure all routing paths are tested.
cg_sideband_user	Coverage of TUSER bit toggling.
cg_wakeup	Coverage of TWAKEUP assertion prior to and during TVALID.
cg_parity	Coverage of parity bit generation for TDATA, TID, TDEST, and TUSER.
cg_reset_sync	Coverage of ACLK edges relative to ARESETn deassertion.

5. Interface Signals

Signal Name	Direction	Width	Description
ACLK	Input	1	Global Clock Signal. All signals sampled on rising edge.
ARESETn	Input	1	Active-LOW Synchronous Reset. TVALID must be LOW during reset.
TVALID	Output	1	Indicates Master is driving a valid transfer.
TREADY	Input	1	Indicates Slave can accept a transfer.
TDATA	Output	TDATA_WIDTH	Primary data payload.
TSTRB	Output	TDATA_WIDTH /8	Byte qualifier for data/position bytes.
TKEEP	Output	TDATA_WIDTH /8	Byte qualifier for null byte removal.
TLAST	Output	1	Indicates end of packet boundary.
TID	Output	TID_WIDTH	Stream Source Identifier.
TDEST	Output	TDEST_WIDTH	Course Routing Destination Identifier.
TUSER	Output	TUSER_WIDTH	User-defined sideband information.
TWAKEUP	Output	1	AXI5 Power Wakeup signal.
TDATACHK	Output	TDATA_WIDTH /8	AXI5 Parity bits for TDATA.

6. Verification Environment Structure

axi_stream_master_vip_tb/

Verification Plan Report

```
-- master_agent/
|  |-- axi_stream_master_driver.sv
|  |-- axi_stream_master_monitor.sv
|  |-- axi_stream_master_sequencer.sv
|  |-- axi_stream_master_agent_config.sv
|  |-- axi_stream_master_agent.sv
|  |-- axi_stream_master_callback.sv
|-- env/
|  |-- axi_stream_master_scoreboard.sv
|  |-- axi_stream_master_env_config.sv
|  |-- axi_stream_master_env.sv
|-- sequences/
|  |-- axi_stream_master_base_seq.sv
|  |-- axi_stream_master_burst_seq.sv
|  |-- axi_stream_master_seq_item.sv
|-- top/
|  |-- axi_stream_master_tb_top.sv
|  |-- axi_stream_master_test.sv
|  |-- axi_stream_master_defines.sv
|  |-- axi_stream_master_if.sv
```

7. SystemVerilog Implementation

axi_stream_master_defines.sv (Architecture Blueprint)

COMPONENT: axi_stream_master_defines

EXTENDS: N/A

MEMBERS:

- parameter TDATA_WIDTH = 32
- parameter TID_WIDTH = 8
- parameter TDEST_WIDTH = 4
- parameter TUSER_WIDTH = 4
- parameter HAS_PARITY = 1
- parameter HAS_WAKEUP = 1

LOGIC:

- Centralized global parameters to ensure interface-to-transaction width consistency.

axi_stream_master_seq_item.sv (Architecture Blueprint)

COMPONENT: axi_stream_master_seq_item

EXTENDS: uvm_sequence_item

MEMBERS:

- rand [bit[TDATA_WIDTH-1:0]] tdata
- rand [bit[TDATA_WIDTH/8-1:0]] tkeep
- rand [bit[TDATA_WIDTH/8-1:0]] tstrb
- rand [bit] tlast
- rand [bit[TID_WIDTH-1:0]] tid
- rand [bit[TDEST_WIDTH-1:0]] tdest
- rand [bit[TUSER_WIDTH-1:0]] tuser

Verification Plan Report

- rand [int] delay_cycles (Range 0-10)

METHODS & LOGIC:

- do_copy/do_compare: Implementation for deep transaction comparison.
- constraint valid_qualifiers: TKEEP & !TSTRB == Reserved (Forbidden).

axi_stream_master_driver.sv (Architecture Blueprint)

COMPONENT: axi_stream_master_driver

EXTENDS: uvm_driver

CONFIG_REF: axi_stream_master_agent_config

MEMBERS:

- virtual axi_stream_if vif

METHODS & LOGIC:

- run_phase:
 1. Reset Handling: Wait for ARESETn=1. Drive TVALID=0.
 2. Main Loop:
 - Get item from sequencer.
 - If HAS_WAKEUP, assert TWAKEUP, wait config.latency.
 - Drive TDATA, TKEEP, TSTRB, TLAST, TID, TDEST, TUSER.
 - Calculated Odd Parity and drive TDATACHK.
 - Assert TVALID.
 - Wait for rising edge of ACLK where VIF.TREADY == 1.
 - Deassert TVALID (or hold for next beat).
 - Signal item_done.

axi_stream_master_monitor.sv (Architecture Blueprint)

COMPONENT: axi_stream_master_monitor

EXTENDS: uvm_monitor

MEMBERS:

- virtual axi_stream_if vif
- uvm_analysis_port #(axi_stream_master_seq_item) ap

METHODS & LOGIC:

- run_phase:
 - Forever Loop:
 - Sample signals on ACLK rising edge.
 - If TVALID == 1 and TREADY == 1 (Handshake):
 - Create seq_item.
 - Populate item from physical pins.
 - Perform immediate parity check (if enabled).
 - Write to analysis port.
 - Check for REQ_MST_01 violation: If TVALID falls without handshake.

axi_stream_master_scoreboard.sv (Architecture Blueprint)

COMPONENT: axi_stream_master_scoreboard

EXTENDS: uvm_scoreboard

MEMBERS:

- uvm_tlm_analysis_fifo #(axi_stream_master_seq_item) master_fifo

Verification Plan Report

- bit [7:0] local_ref_mem [int]

METHODS & LOGIC:

- write_master: Function to capture and store expected data in association with TID.
- check_phase: Compare local reference model against monitored Slave (monitored via slave_fifo - out of scope for this plan).
- Focus on REQ_MST_03 (Integrity) and REQ_MST_05 (TLAST packet markers).